



King AbdulAziz University

Faculty of Computing & Information Technology

Computer Science Department

Course: CPCS204 (Data Structures)

Term: 3rd Semester (Spring 2023)

Final Exam (Thursday June 08, 2023) (10:30 am – 12:30 pm)

Time: 120 minutes

without SOLUTION

Student ID: _____

Section: _____

Student Name: _____

Instructor: _____

Underline Your Instructor:

Dr. Muhammad Umair	Dr. Abdullah Marish
Dr. Ali Nasir	Dr. Waseem Abdul Qayyum
Dr. Asaad Ahmed	Dr. Abdul Mohsen

Q#	SO	Marks		Q#	SO	Marks	
1	02	20		5	02	15	
2	02	20		6	01	10	
3	02	20		7	01	10	
4	02	05					

Question#1:

(SO 02. Linked Lists _____ /20)

Suppose a non-empty single link list contains **even values** only. Write a method which receives two parameters i.e. first **head** of the link list and second an integer **value**. The method should find the value in the list and if found then **insert** TWO nodes; one as **predecessor** and other as a **successor** to the node containing the **value** passed as parameter. **Predecessor** must have "**value - 1**" and successor must have "**value+1**" in their respective data member.

Example:

Value Passed: 68

Initial List:

10 → 12 → 30 → 40 → 68 → 74 → 82 → 86

Updated List:10 → 12 → 30 → 40 → **67** → 68 → **69** → 74 → 82 → 86

```
public static void exam01 (LLnode head, int value) {
```

(08)

(04)

(08)

}

Question#2:

(SO 02. Recursion _____ /20)

The Fibonacci sequence is a famous bit of mathematics, and it happens to have a recursive definition. The **first two values** in the sequence are **0 and 1**. Each subsequent value is the sum of the previous two values, so the whole sequence is: 0, 1, 1, 2, 3, 5, 8, 13, 21 and so on. Write a recursive method i.e. **exam02(n)** which returns the **nth Fibonacci number**, for the given sequence. Moreover, give the answer of the followings and trace your algorithm for **exam02 (6)**

Short Answers:

Base Case 01: -----n = -----, Return:----- (01)

Base Case 02: -----n = -----, Return:----- (01)

Generic Formula: - (02)

Code:

```
public static int exam02 (int n) {
```

(04)

Tracing:

(12)

Question# 3:

(SO 02. Binary Search Trees: _____ /20)

Construct a BST for the following values. You are required to show **only the final tree**. No need to show the intermediate steps. Once the BST constructed then write the output of traversal of the tree using **INORDER** and **BFS**

GEN, INT, FIN, CON, VAL, OUT, BST, BIN, TRA, FOL, TRE, DFS, THE, INO, SHO

Hint:

A	B	C	D	E	F	G	H	I	J	K	L	M
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Constructed BST:

(15)

INORDER:

(2.5)

BFS:

(2.5)

Question# 4:

(SO 02. Algorithm Analysis: _____ /05)

Analyze the following method and write its **Big - O approximation**. Also **justify** your answer.

NOTE: You will get only **ONE point** for correct answer without appropriate explanation.

```
public static int exam04 (int n) {
    int x=5;
    int k= n;
    for (i=1; i<=k; i++) {
        while (n >0) {
            x = x*2;
            n = n/2;
        }
        n = k;
    }

    int i=0, j=0;
    while (i < n)
        while (j < n)
            j++;
        i++;
    }
}
```

Big - O Approximation:

$T(n) = \text{-----} O (\quad) \text{-----}$ (1)

Explanation:

(4)

Question# 5:

(SO 02. Stack: ____ /15)

Convert the given infix expression to a Postfix expression. Postfix expression is given for the verification. Once the postfix expression is converted then evaluate it using a Stack.

A
B
C

34 + 6 - (2 * 5 - (3 + 43 / 5 % (10 - 6) + 13 - 8) * 7) + 9

A is after - and before (

B is after % and before (

C is after) and before *

NOTE: you must show the contents of the Stack at the indicated points (A, B, and C) in the postfix expression.

Postfix Conversion:

A

B

C

(9)

Postfix expression: _____

Postfix Evaluation:

A
B
C

34 6 + 2 5 * 3 43 5 / 10 6 - % + 13 + 8 - 7 * - - 9 +

A is after * and before 3

B is after % and before +

C is after - and before -

A

B

C

(6)

Expression Value: _____

Question# 6:

(SO 01. Sorting Algorithms: _____ /10)

Given the following list of numbers [43, 76, 13, 63, 27, 34, 17, 8, 51, 9], show the contents of the list after every iteration according to the **Bubble Sort** algorithm to arrange the values in a **descending order**?

	43	76	13	63	27	34	17	8	51	9
1 st										
2 nd										
3 rd										
4 th										
5 th										
6 th										
7 th										
8 th										
9 th										
10 th										
Final	76	63	51	43	34	27	17	13	9	8

Question#7:

(SO 01. Heap Tree _____ /10)

Construct a binary **Max-Heap** Tree by inserting the following numbers one by one.

Branch, Level, Height, Nodes, Tree, Heap, Max, Build, Greater, Smaller

Hint:

A	B	C	D	E	F	G	H	I	J	K	L	M
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Max Heap: